

Practical Assignment:

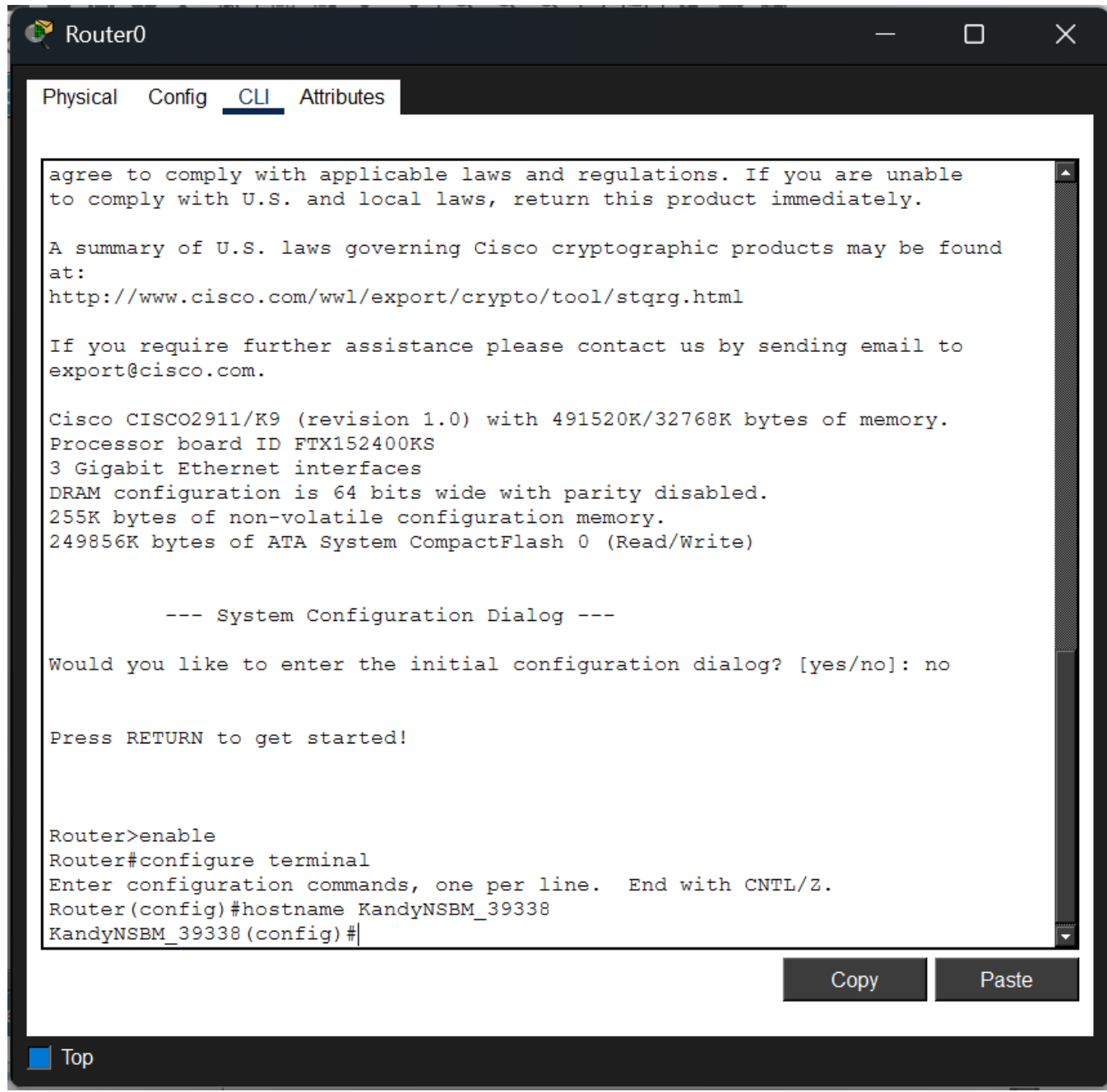
Router Configuration and
Routing Protocols using
Cisco Packet

Module:
Computer Networks

Diwyanjalee P D C

39338

Task 1 – Basic Router Configuration



The screenshot shows a web-based interface for configuring a Cisco router named Router0. The interface has a dark header with the router name and standard window controls. Below the header, there are four tabs: Physical, Config, CLI (which is selected and underlined), and Attributes. The main content area displays the following text:

```
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found
at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname KandyNSBM_39338
KandyNSBM_39338(config)#
```

At the bottom right of the main content area, there are two buttons: "Copy" and "Paste". At the bottom left of the interface, there is a blue square icon followed by the text "Top".

```
at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname KandyNSBM_39338
KandyNSBM_39338(config)#line console 0
KandyNSBM_39338(config-line)#password NSBM
KandyNSBM_39338(config-line)#login
KandyNSBM_39338(config-line)#exit
KandyNSBM_39338(config)#
```

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Task 2 – Static Routing

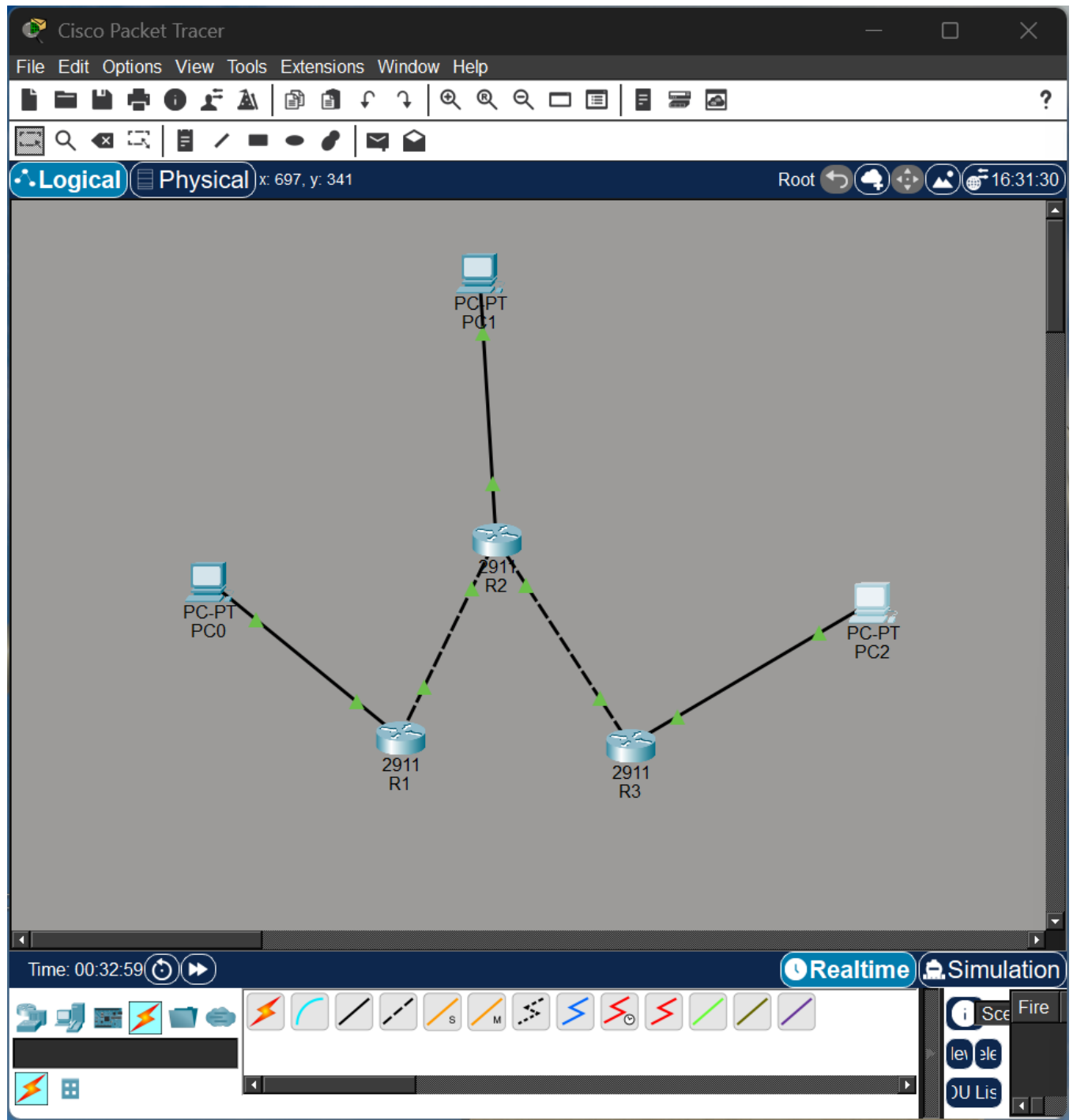
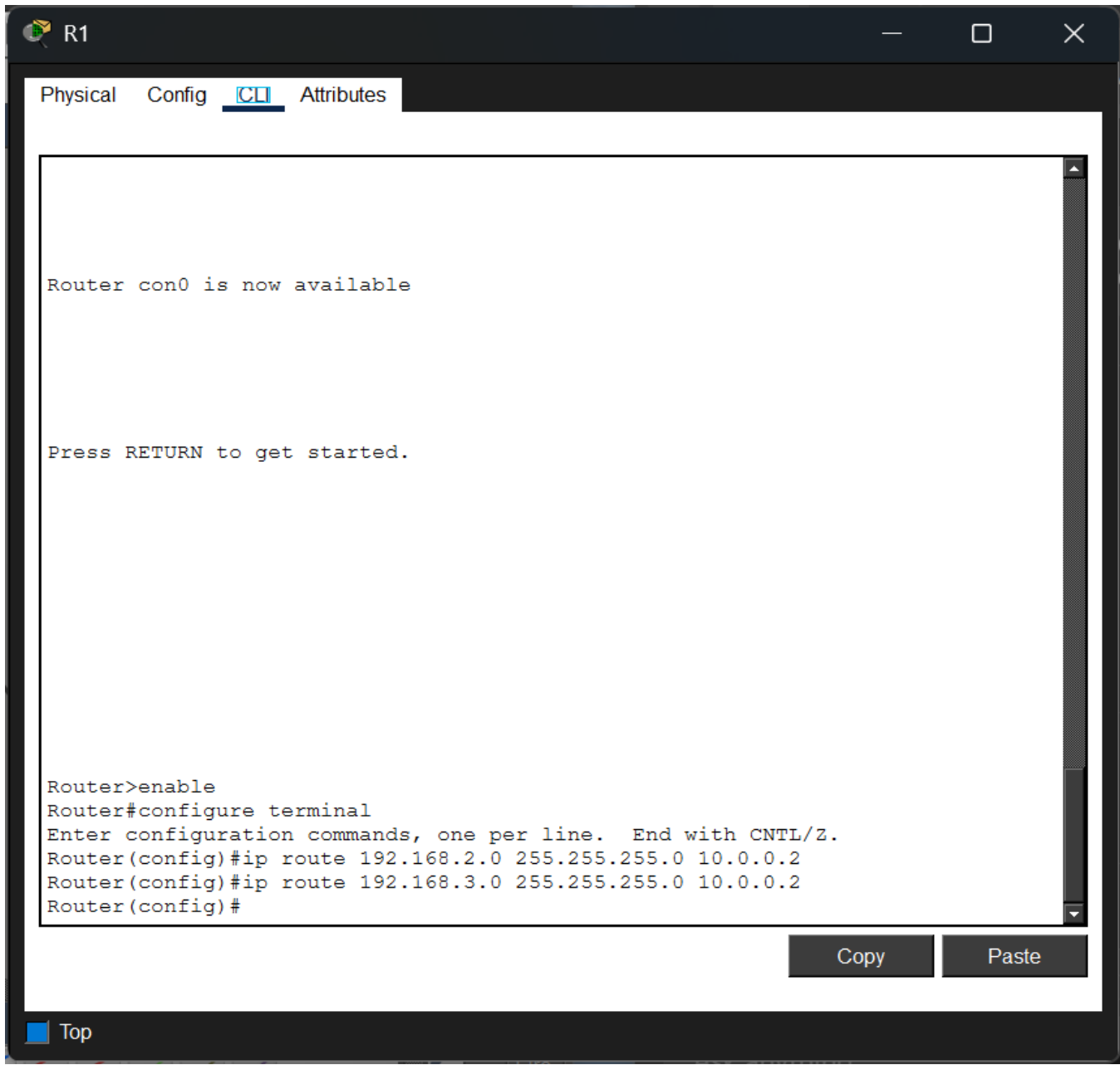


Figure 1 Network diagram with IP addresses

Configuration commands used on each router



Router con0 is now available

Press RETURN to get started.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1

%Invalid next hop address (it's this router)

Router(config)#ip route 192.168.3.0 255.255.255.0 20.0.0.2

Router(config)#

[Copy](#)[Paste](#)

Router con0 is now available

Press RETURN to get started.

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 192.168.1.0 255.255.255.0 20.0.0.1

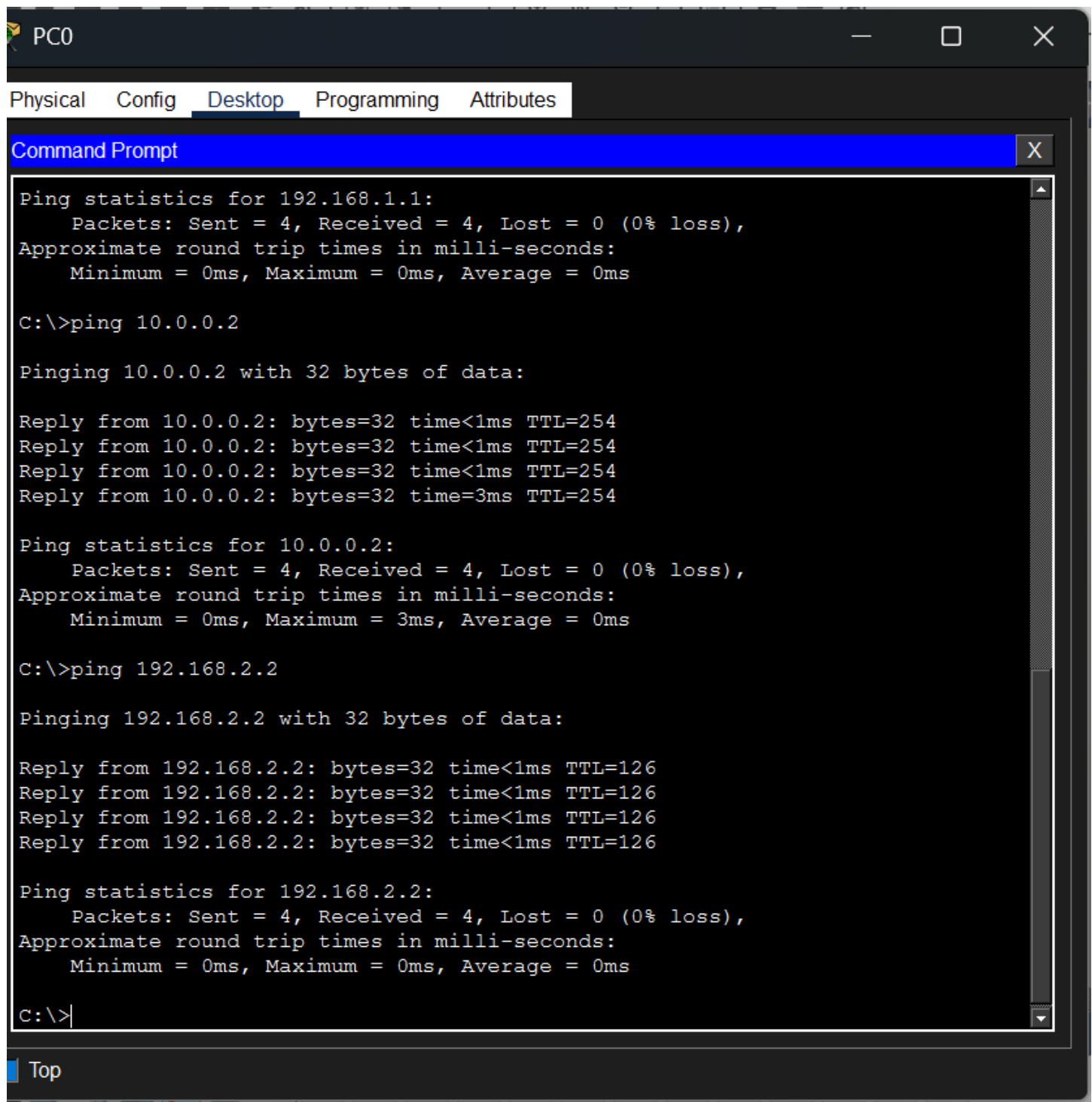
Router(config)#ip route 192.168.2.0 255.255.255.0 20.0.0.1

Router(config)#

Copy

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Ping test screenshots



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of two ping tests. The first test is for 192.168.1.1, showing 4 packets sent and received with 0% loss and 0ms round trip times. The second test is for 10.0.0.2, showing 4 packets sent and received with 0% loss and 0ms round trip times. The third test is for 192.168.2.2, showing 4 packets sent and received with 0% loss and 0ms round trip times. The Command Prompt window has a blue title bar and a scroll bar on the right. The PC0 window has a dark gray title bar and a light gray border.

```
PC0
Physical Config Desktop Programming Attributes

Command Prompt

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=254
Reply from 10.0.0.2: bytes=32 time<1ms TTL=254
Reply from 10.0.0.2: bytes=32 time<1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=3ms TTL=254

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.2.2

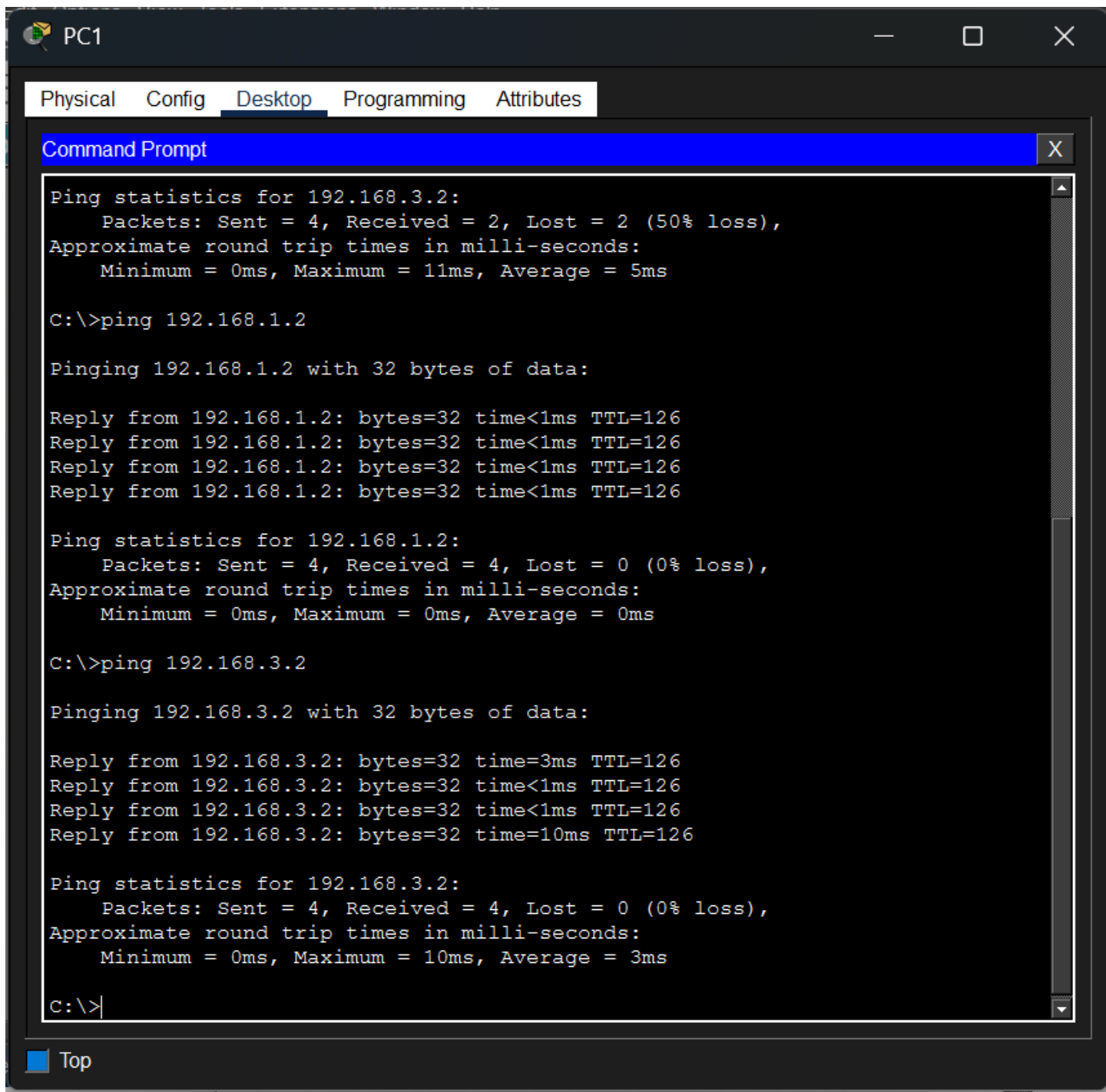
Pinging 192.168.2.2 with 32 bytes of data:

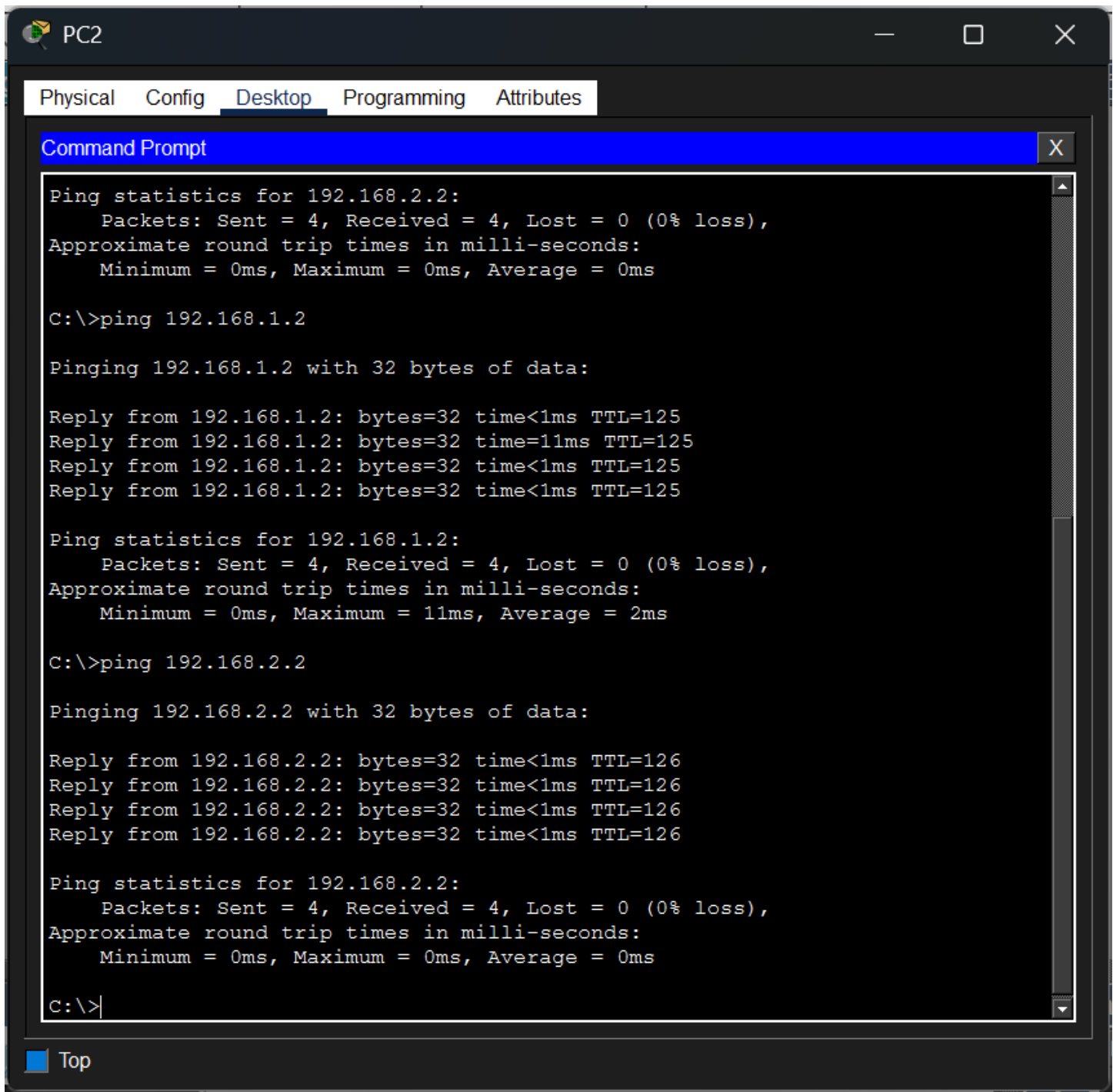
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Top





The screenshot shows a window titled "PC2" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of ping tests to 192.168.2.2 and 192.168.1.2, followed by the execution of ping commands to these addresses. The ping statistics for 192.168.2.2 show 0% loss and 0ms round trip times. The ping statistics for 192.168.1.2 show 0% loss and an average round trip time of 2ms. The Command Prompt also shows the execution of the ping command to 192.168.1.2, which shows 0% loss and an average round trip time of 2ms. The Command Prompt is currently at the prompt "C:\>".

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time=11ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125
Reply from 192.168.1.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

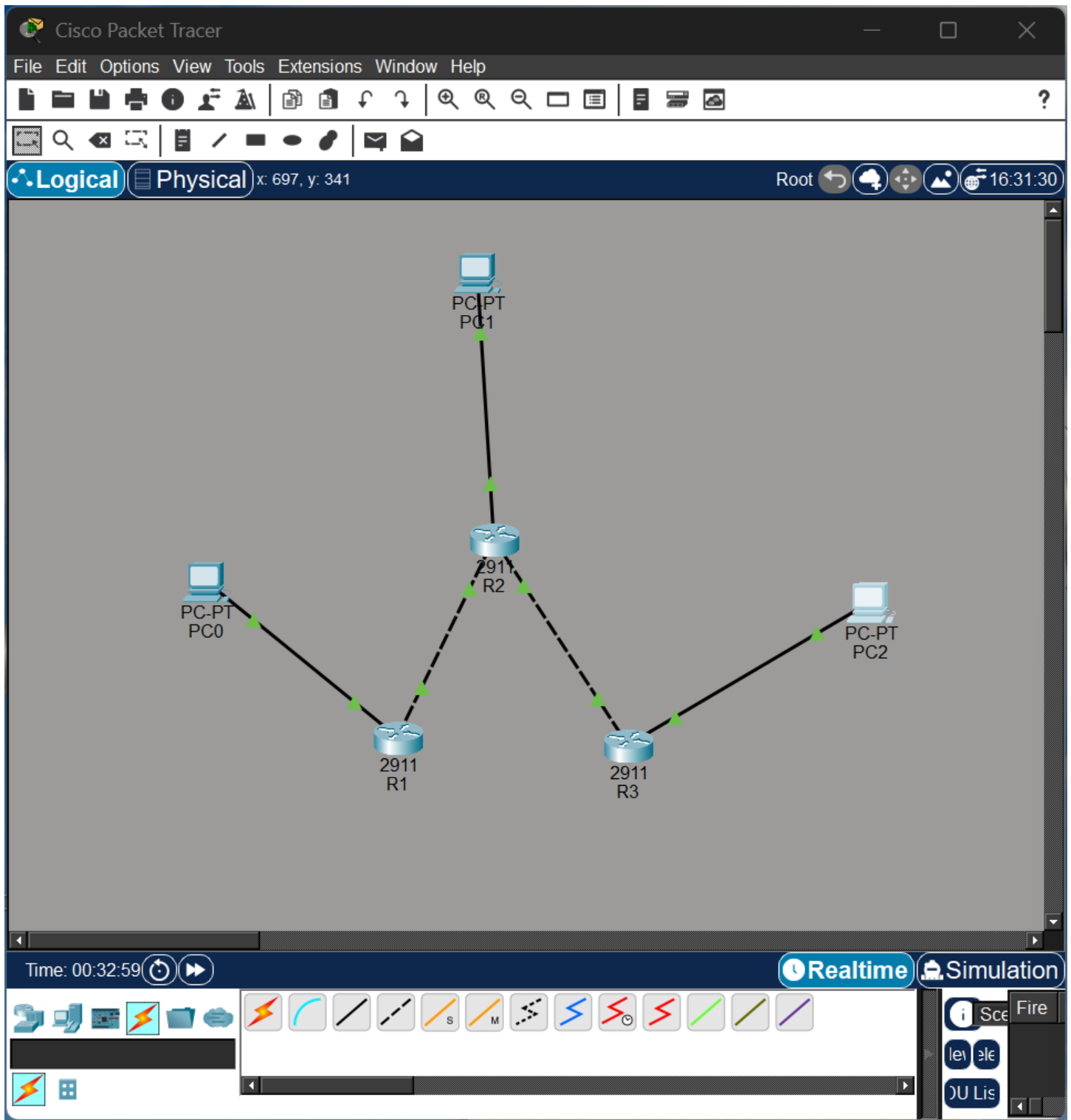
Brief Explanation on Static Routing Configuration

In this task, three routers and three separate networks were configured using IP addressing and static routing. Static routes were manually added on each router using the ip route command to allow communication between different networks. Connectivity between all networks was verified successfully using ping tests. Static routing provides fixed paths for data transmission and is suitable for small networks.

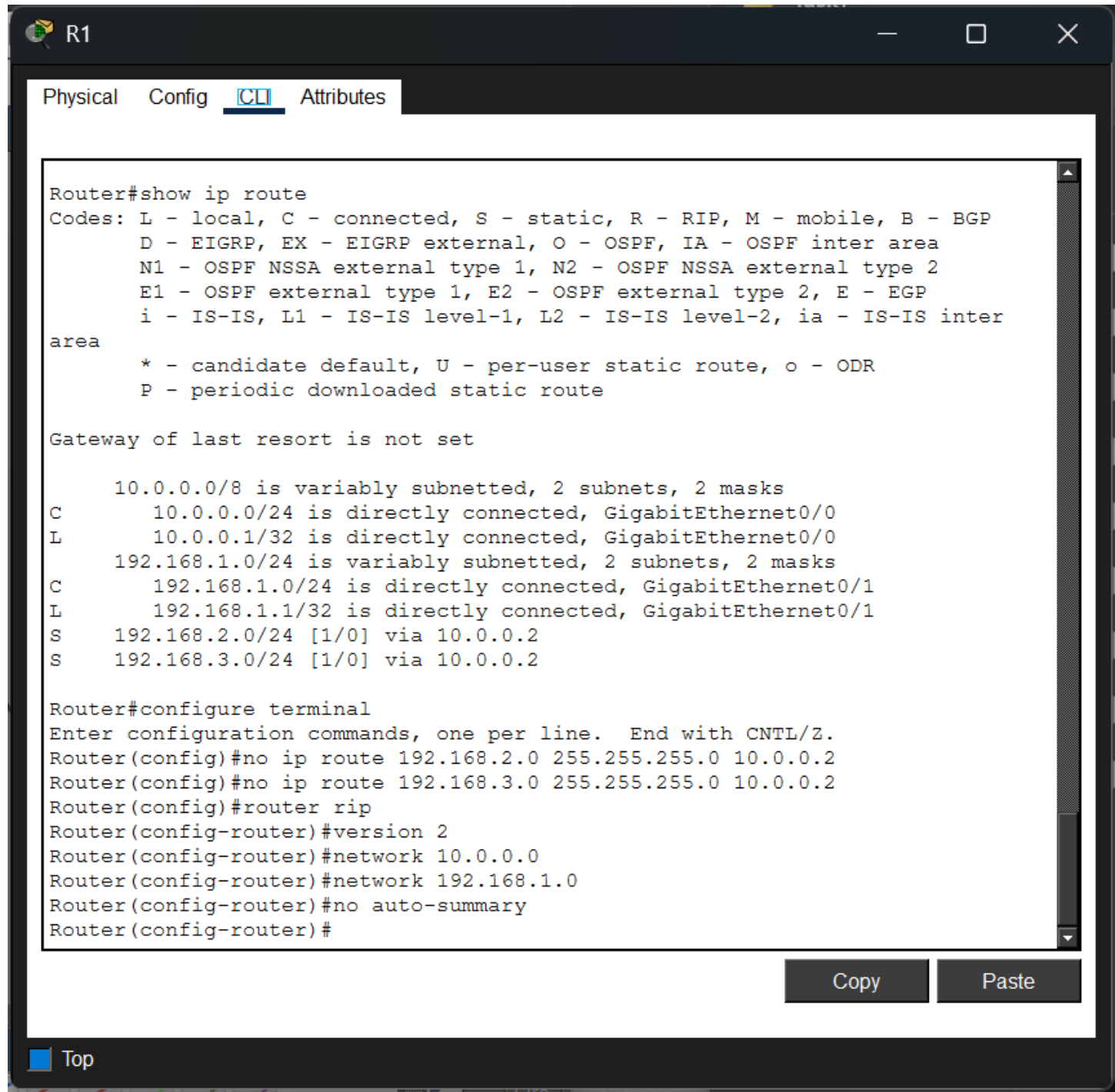
Task 3 – Dynamic Routing (RIP and EIGRP)

1. Network diagram

- Used the same diagram by clearing the before routes.



2. Configuration commands



R1

Physical Config **CLI** Attributes

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
  192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
S       192.168.2.0/24 [1/0] via 10.0.0.2
S       192.168.3.0/24 [1/0] via 10.0.0.2

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#no ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#no auto-summary
Router(config-router)#
```

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Top

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/24 is directly connected, GigabitEthernet0/0
L 10.0.0.2/32 is directly connected, GigabitEthernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/24 is directly connected, GigabitEthernet0/1
L 20.0.0.1/32 is directly connected, GigabitEthernet0/1
S 192.168.1.0/24 [1/0] via 10.0.0.1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/24 is directly connected, GigabitEthernet0/2
L 192.168.2.1/32 is directly connected, GigabitEthernet0/2
S 192.168.3.0/24 [1/0] via 20.0.0.2

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#no ip route 192.168.1.0 255.255.255.0 10.0.0.1

Router(config)#no ip route 192.168.3.0 255.255.255.0 20.0.0.2

Router(config)#router rip

Router(config-router)#version 2

Router(config-router)#network 10.0.0.0

Router(config-router)#network 20.0.0.0

Router(config-router)#network 192.168.2.0

Router(config-router)#no auto-summary

Router(config-router)#

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```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

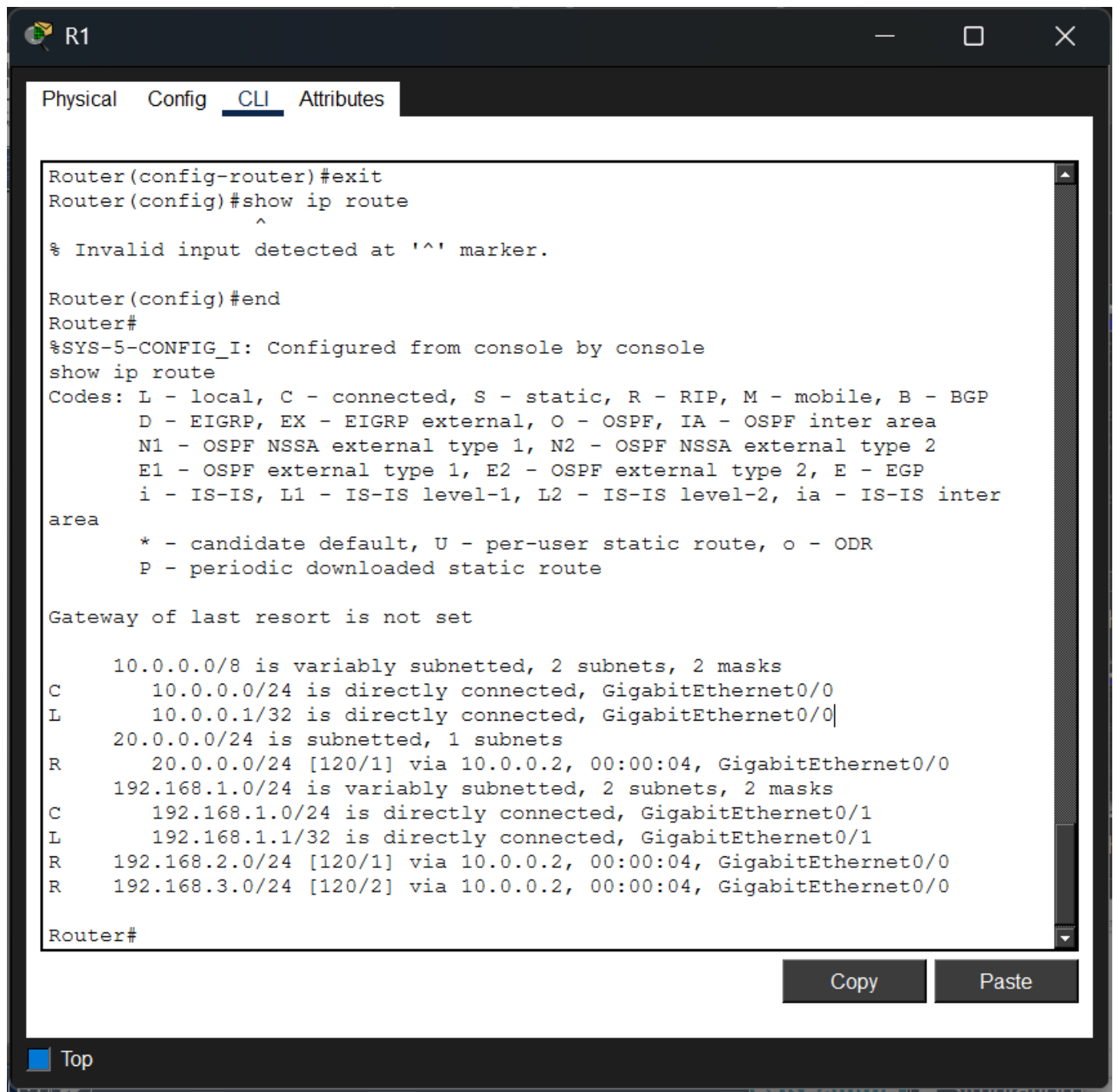
```
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/0
L       20.0.0.2/32 is directly connected, GigabitEthernet0/0
S       192.168.1.0/24 [1/0] via 20.0.0.1
S       192.168.2.0/24 [1/0] via 20.0.0.1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
L       192.168.3.1/32 is directly connected, GigabitEthernet0/1
```

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 192.168.1.0 255.255.255.0 20.0.0.1
Router(config)#no ip route 192.168.2.0 255.255.255.0 20.0.0.1
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 20.0.0.0
Router(config-router)#network 192.168.3.0
Router(config-router)#no auto-summary
Router(config-router)#
```

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3.Screenshots of routing tables



```
Router(config-router)#exit
Router(config)#show ip route
^
% Invalid input detected at '^' marker.

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0|
R       20.0.0.0/24 is subnetted, 1 subnets
R       20.0.0.0/24 [120/1] via 10.0.0.2, 00:00:04, GigabitEthernet0/0
R       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
R       192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:04, GigabitEthernet0/0
R       192.168.3.0/24 [120/2] via 10.0.0.2, 00:00:04, GigabitEthernet0/0

Router#
```

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Top


```
Router(config-router)#network 10.0.0.0
Router(config-router)#network 20.0.0.0
Router(config-router)#network 192.168.2.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
R       192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:09, GigabitEthernet0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/2
L       192.168.2.1/32 is directly connected, GigabitEthernet0/2
R       192.168.3.0/24 [120/1] via 20.0.0.2, 00:00:09, GigabitEthernet0/1

Router#
```

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```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 20.0.0.0
Router(config-router)#network 192.168.3.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

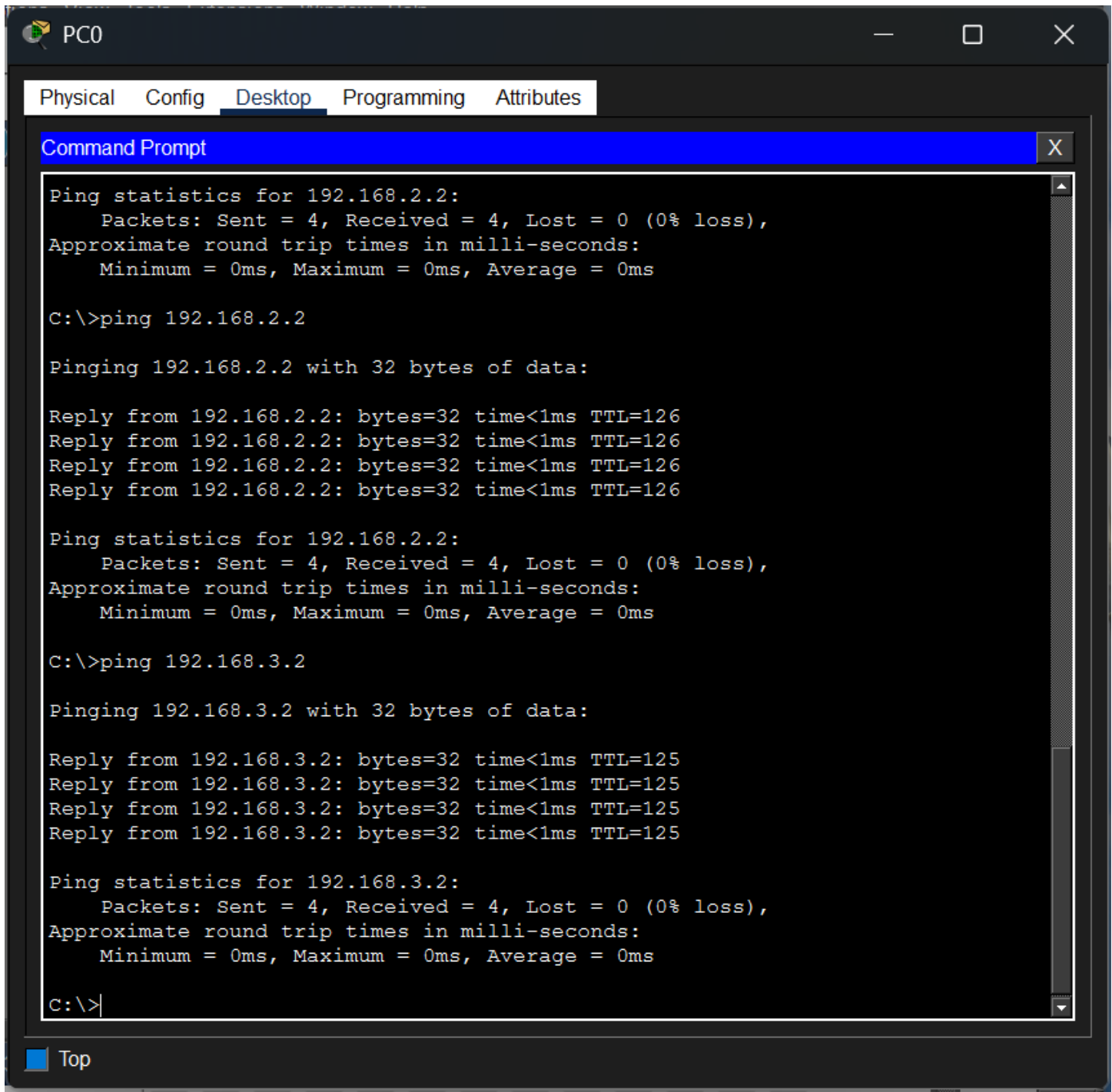
    10.0.0.0/24 is subnetted, 1 subnets
R       10.0.0.0/24 [120/1] via 20.0.0.1, 00:00:19, GigabitEthernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/0
L       20.0.0.2/32 is directly connected, GigabitEthernet0/0
R       192.168.1.0/24 [120/2] via 20.0.0.1, 00:00:19, GigabitEthernet0/0
R       192.168.2.0/24 [120/1] via 20.0.0.1, 00:00:19, GigabitEthernet0/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
L       192.168.3.1/32 is directly connected, GigabitEthernet0/1

Router#
```

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Paste

4. Ping test results



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of two ping tests. The first test is for 192.168.2.2, showing 4 packets sent, 4 received, 0% loss, and 0ms round trip times. The second test is for 192.168.3.2, also showing 4 packets sent, 4 received, 0% loss, and 0ms round trip times. The Command Prompt window has a blue title bar and a scroll bar on the right. A "Top" button is visible at the bottom left of the window.

```
PC0
Physical Config Desktop Programming Attributes

Command Prompt

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.2

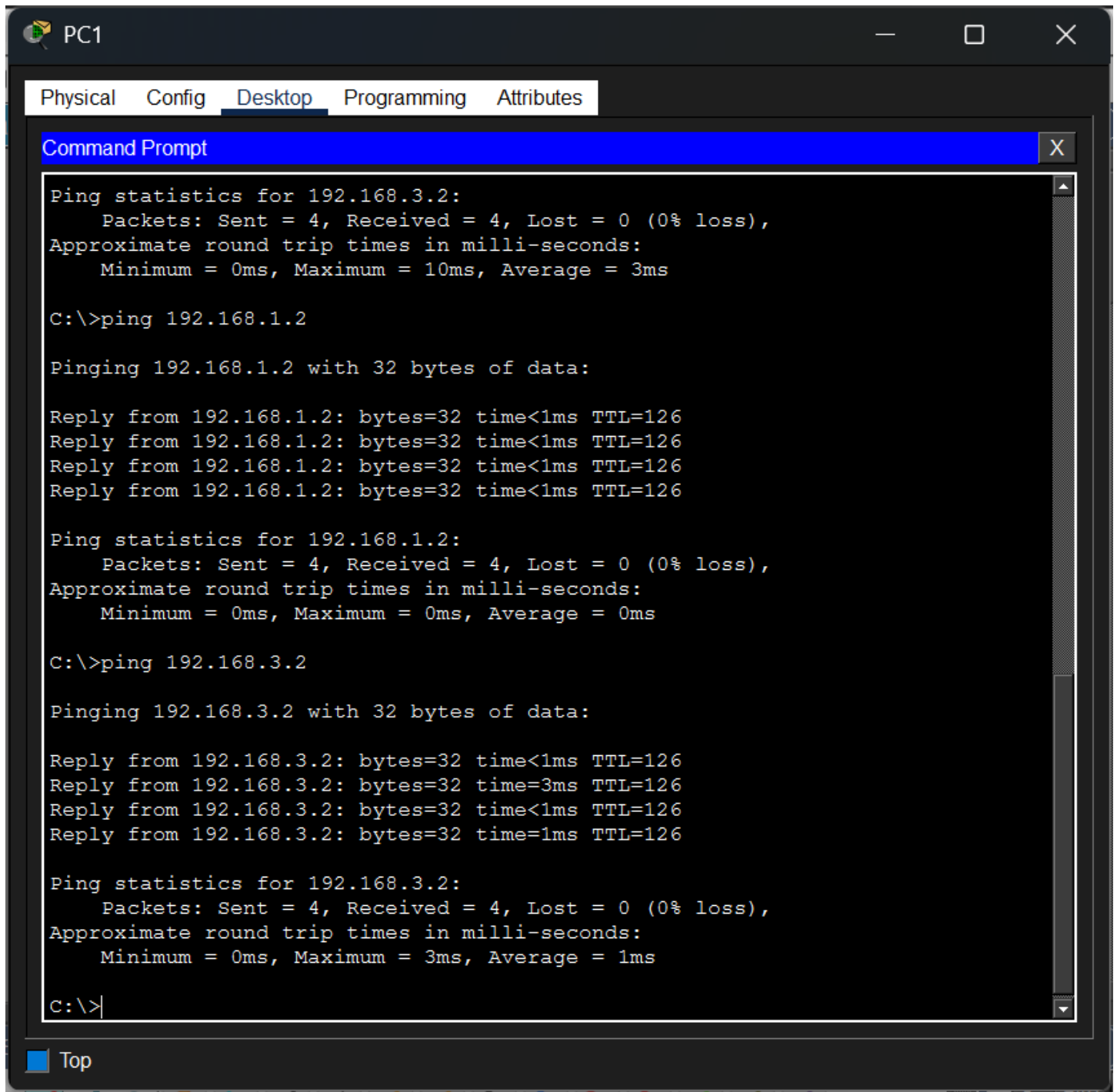
Pinging 192.168.3.2 with 32 bytes of data:

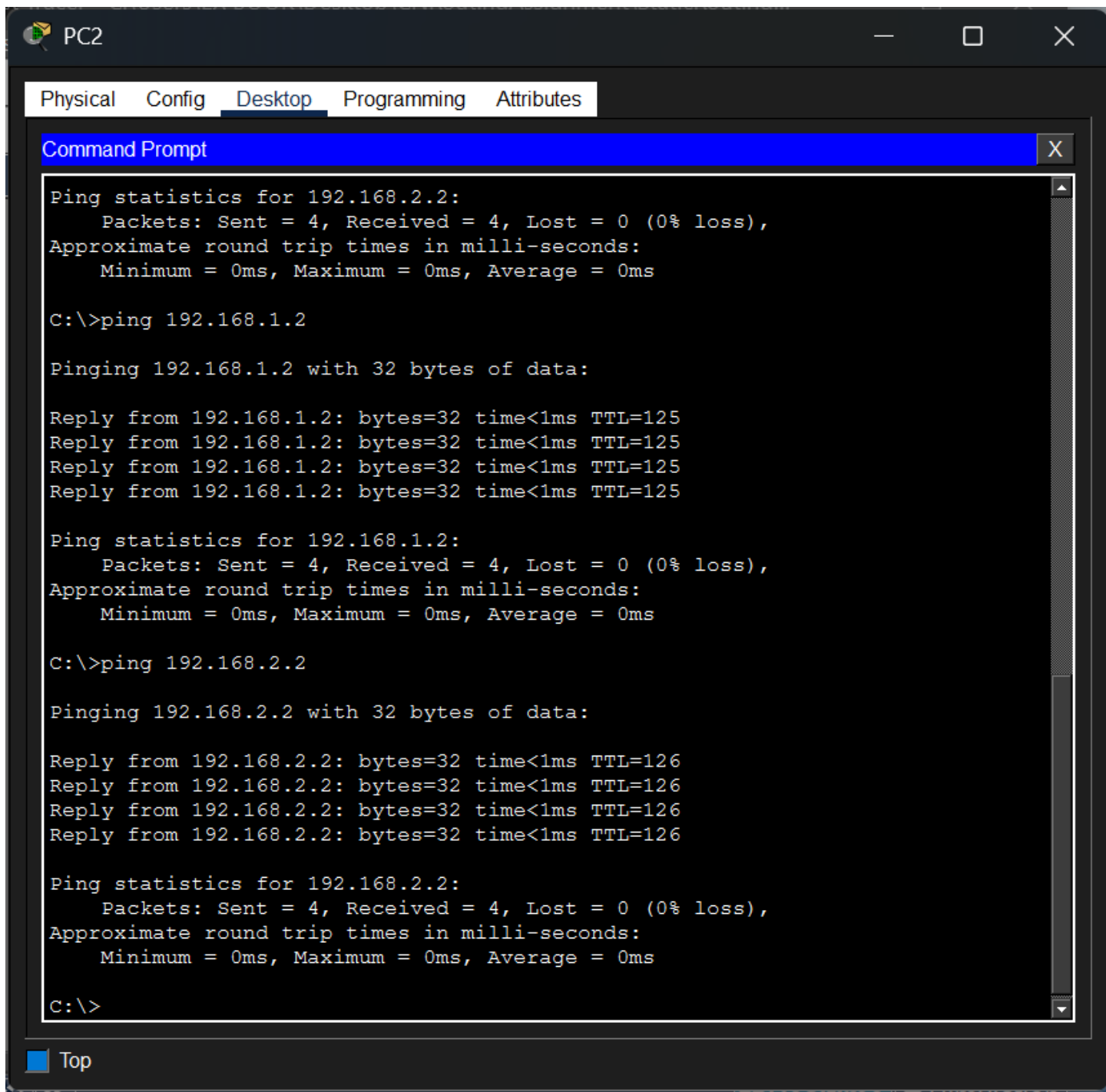
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

Top
```





Screenshots Of EGIRP Configuration Commands And Routing Tables

 R1

Physical Config CLI Attributes

```
R      192.168.3.0/24 [120/2] via 10.0.0.2, 00:00:04, GigabitEthernet0/0

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no router rip
Router(config)#router eigrp 100
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1

Router#
```

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```
Router(config-router)#no auto-summary
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 20.0.0.2 (GigabitEthernet0/1) is
up: new adjacency

Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/2
L       192.168.2.1/32 is directly connected, GigabitEthernet0/2
D       192.168.3.0/24 [90/28416] via 20.0.0.2, 00:00:57, GigabitEthernet0/1

Router#
```

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```
Router(config)#no router rip
Router(config)#router eigrp 100
Router(config-router)#network 20.0.0.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 20.0.0.1 (GigabitEthernet0/0) is
up: new adjacency

Router(config-router)#network 192.168.3.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/24 is directly connected, GigabitEthernet0/0
L       20.0.0.2/32 is directly connected, GigabitEthernet0/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
L       192.168.3.1/32 is directly connected, GigabitEthernet0/1

Router#
```

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Brief Explanation on RIP Routing

In this task, RIP (Routing Information Protocol) was configured on all routers to enable dynamic routing. RIP automatically exchanges routing information between routers and updates routing tables dynamically. The router `rip`, version 2, and network commands were used to advertise connected networks. Successful communication between all networks was verified using ping tests and routing table entries marked with R.

Brief Explanation on EIGRP Routing

In this task, the RIP configuration was removed and EIGRP (Enhanced Interior Gateway Routing Protocol) was configured using Autonomous System number 100. EIGRP dynamically shares routing information and provides faster convergence and better performance compared to RIP. The router `eigrp 100` and network commands were used to advertise networks. Successful communication between all networks was verified using ping tests and routing table entries marked with D.

Conclusion

This assignment successfully demonstrated the configuration and implementation of router networking concepts using Cisco Packet Tracer. Basic router configuration was performed by setting hostnames, console passwords, and login authentication to improve device security and management. Static routing was configured to establish communication between multiple networks using manually defined routes. In addition, dynamic routing protocols RIP and EIGRP were implemented to automatically exchange routing information between routers. Connectivity between all networks was successfully verified through ping tests and routing table analysis. Overall, this assignment improved practical knowledge of router configuration, IP addressing, static routing, and dynamic routing protocols in computer networks.